

PAPER_1623 — Lunar Synodic Month = $T_{\text{synod}} = D_{\text{crit}} + D - F \cdot D - F \cdot \Phi + F^2 \cdot K = 29.5375$ days

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Astronomy **Date:** June 18, 2026 **Status:** CLOSED — 0.025% closure

Observation

PAPER_1209Z_S582 (Astronomy Unified Proof Set) gives the formula:

$$T_{\text{synod}} = D_{\text{crit}} + D - F \cdot D - F \cdot \Phi + F^2 \cdot K = 29.5375 \text{ days}$$

Residual: **0.025%**.

UQFF Closed Identity

$$T_{\text{synod}} = D_{\text{crit}} + D - F \cdot D - F \cdot \Phi + F^2 \cdot K = 29.5375 \text{ days} \quad 0.025\%$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Empirical astronomy measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209Z_S582
- Related: PAPER_1494-1615 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "synodic_month_29_53"})`

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, June 18, 2026, Youngstown OH.